

Water EC/TDS transmitter Manual

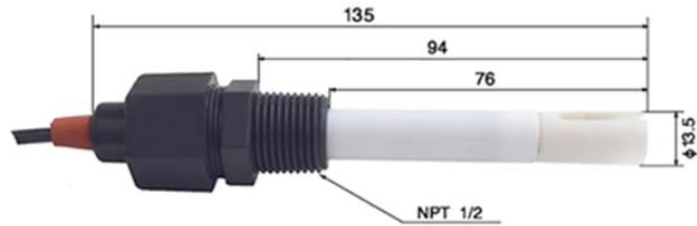
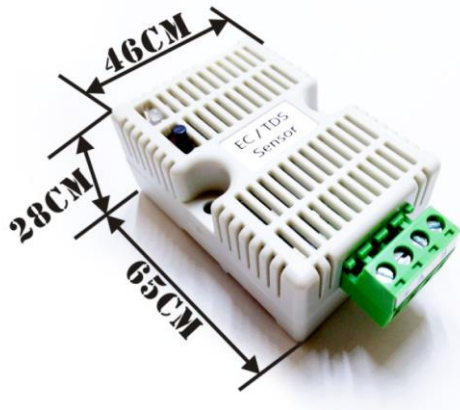


EC - TDS transmitter has a key to on-line calibration, automatic temperature compensation, calibration electrode quality alarm, drop the electricity protection (can make the calibration results and preset data is not lost due to power off or power failure), over-current protection, overvoltage protection, high measuring accuracy, fast response, long life and other characteristics. Products small size, light weight, easy to install and maintain, standard industrial signal output (0, 0 to 10 v to 5 v, 4-20 ma, ModbusRTU485) can satisfy the connection of real-time monitoring equipment. Convenient products and all kinds of control equipment, display instrument connection, realize the EC, TDS status online monitoring. This product is suitable for: electroplating, fermentation, food processing, wastewater treatment, metallurgy, environmental protection, etc.

Specification

Items	Parameters
EC range	Options: 0-440 us/cm、0-4400 us/cm、0-44000 us/cm
EC precision	$\leq \pm 2\%$ F.S
TDS range	Options: 0-200PPM、0-2000PPM、0-20000PPM
TDS precision	$\leq \pm 2\%$ F.S
Power supply	DC 12V-24V
Output	0-5V / 0-10V / 4-20mA / RS485
load resistance	Voltage output: R load $\geq 5K \Omega$ Current output: R load $\leq (U_{vcc}-3)/0.02\Omega$
Consumption	<1W
Storage condition	10-50℃ (-20-+80C peak) 20-60%RH
Working environment	0~100℃
dimension	65 X 46 X 28.5mm ³

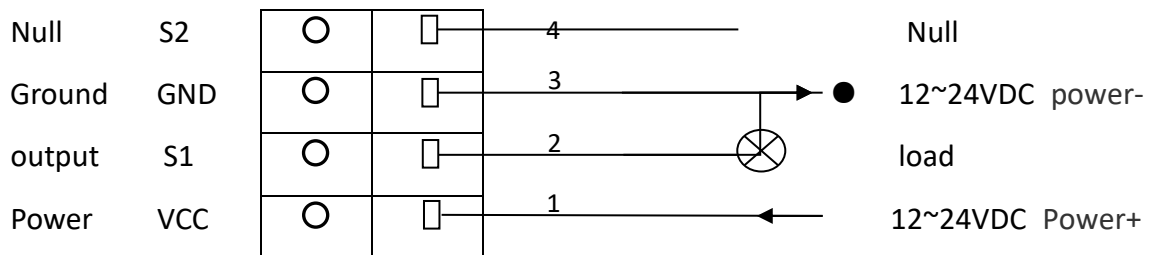
Size



ABS conductivity electrode with 5m cable
 constant: K=1.0
 Temperature: 0-50°C pressure: 0-0.5MPa

Wiring

1. Analog output:



The linear relation between analog signals and the corresponding physical quantities:

0-5 v output: quantity = corresponding range x sensor output voltage / 5.0

0-10 v output: quantity = corresponding range x sensor output voltage / 10.0

4-20 ma output: quantity = corresponding range x (sensor output current-4.0) / 16.0

For example: 0-5 v output, range 4400 us/cm, if the sensor output voltage is 3v

EC = 4400/5.0 us/cm x sensor output voltage

= 4400 us/cm x 3.0/5.0 = 2640 us/cm

For example: 4-20 mA output, range 2000 PPM, if the sensor output current is 12mA

TDS = 2000 PPM x (sensor output current-4.0) / 16.0

= 2000 PPM x (12.0-4.0) / 16.0 = 1000 PPM

2. RS485 output

Wiring

1	+	DC 12-24V power+
2	A	RS485+ (A+)
3	-	DC 12-24V power-
4	B	RS485- (B-)

RS485 communication Default parameters: 9600,n,8,1

Default device address is 1

Modbus RTU protocol

Read EC/TDS

Register map						
Register address (Hex)	PLC Address (decimal)	meaning	Number of bytes	Function code	R/W	Scale
0000	40001	Value of calibration solution	2	03	Read	0.1
0001	40002	EC/TDS	2	03	Read	0.1
0002	40003	Slave ID	2	03/06	Read/write	

master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x00	0x00	0x02	0xC4	0X0B

Sensor responds

Address	Function Code	Number of byte	Value of calibration solution	EC/TDS value	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x04	0x37 0x32	0xA8 0x57	0x6B	0xB6

Value of calibration solution=3732H/10=14130/10=1314

EC/TDS=A857H/10=43095/10=4309.5

Note: 0-44000uS/cm range, the scale is 1

Set slave ID, master sends

Address	Function Code	Start Address	Number of Points	Number of bytes	ID	Error Check
0~247	10H	0002H	0001H	02H	0001H~00F7H	

Transducer calibration and electrode quality detection

- 1, the electrode with distilled water without CO₂ dry clean and soft tissues.
- 2, correctly connected transmitter power and electrode.
- 3, long press calibration button until the indicator light yellow, then loosen the button.
- 4, fast calibration button twice calibration until the red light flashing, the transmitter into the zero calibration process
(this process is 30 seconds long), after the calibration light is normally on red.
- 5, calibration indicator light red into the prepared standard liquid after normally on electrodes, different range and the corresponding standard fluid is as follows:
 0-440 us range corresponding to the standard liquid: 146.5 us/cm
 A scale of 0-4400 to us and the corresponding standard fluid: 0-2000 parts per million (PPM)
 range 1413 us/cm
 0-44000 us range corresponding to the standard liquid: 12.88 mS/cm
- 6, fast calibration button once the calibration indicator light green flashing, transmitter into the calibration standard process
(this process is 30 seconds long), after the calibration indicator light green normally on 20 seconds, and then prompt the yellow flashing 20 seconds to save the calibration data.
- 7, electrode quality inspection (automatic) and data:
 If electrode damage in the process of calibration, electrode do not match, the calibration red and green lights flashing alarm, flashing lights extinguished, 20 seconds after the end of the calibration, the calibration data is not saved.
 If the electrode trouble-free, during the process of calibration transmitter automatically stored at this time of the calibration data, calibration yellow light is normally on 20 seconds, the end of the calibration.